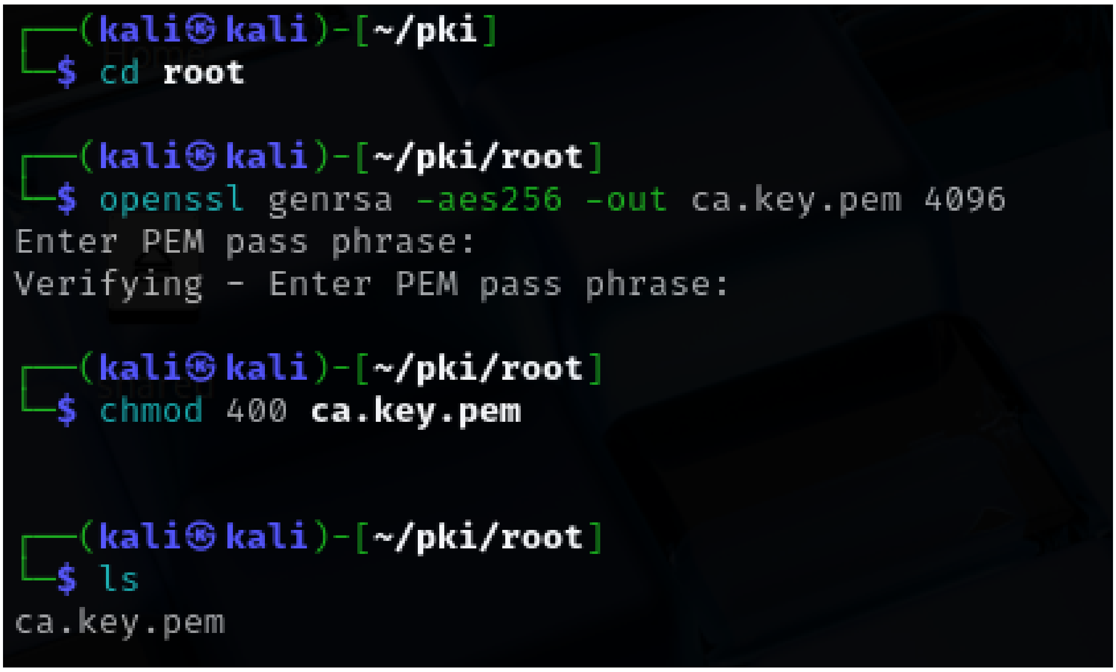
Création de l'environnement de travail :

mkdir -p ~/pki/{root,intermediate,certs}

Création de la CA racine :

Dans root je créer une clé aes256 ,il s'agit de la clé privé de la racine. j'utilise comme passphrase **alexis**
openssl genrsa -aes256 -out ca.key.pem 4096
chmod 400 ca.key.pem



Voici la clé :

```
┌──(kali㉿kali)-[~/pki/root]
└─$ ls
ca.key.pem

┌──(kali㉿kali)-[~/pki/root]
└─$ cat ca.key.pem
-----BEGIN ENCRYPTED PRIVATE KEY-----
MIIJtTBfBgkqhkiG9w0BBQ0wUjAxBgkqhkiG9w0BBQwwJAQQfdHLYowhWD1w6pN7
y2HHbQICCAAwDAYIKoZIhvcNAgkFADAdBglghkgBZQMEASoEECgpmaOmlboI+TEm
jWNPstUEgglQavZkQW9xMxiWOHCKzNjevfF0rV7u6nqFhFDddAyJyeQEwUGogCYa
1r/BR7DgHOooAD71/0ArLwOZfXAIyzqEMA1NPPN4ohionkAqXw0d5eXGhH++S126
ZadOAlzJ5+cF/AhUpw8G/bRl5LivDge4ousySjrJeUPBWtKEGEduODvb6TXbiWza
3M8Hc2in1Ypsru2a2bmnHSV8EeN3aSZeUVEY6KIA/AXHt4pxQNQHvQe88R3nXiA3
LMa7U38LYyG3dXcbL+snPia4Kk8jynkIe9HxgU/2KA6A7DfcHwdcsl2UlbC7aXLz
U0lbySnvRd3Vk3arUzvCNVzw4b7Gr3DC1M+3ICKooCfE80iGpUJb2AmuXs4AZuaZ
CVZJzuRvj+bZt6bpKhqwDb8GJj08sTQhNTOGqLrDsOmTssIoW3p2lKLvrFQCQKGJ
rUJjWxdB5M+icKGLgRE8EmsAs7hSsXKgKQJlMdaINLE07T7DUhX3RfRWfVatl4/H
4oaUYmuf241iz04U2WF941CJtTu+1uFzAgB2iVK37zYtnPgKLjIRYkGBDB4l7cwU
m+IZ70ogVC5vaX+rOf/UqsrJNqo11YMiBDZ8pFsseLgumCf9f74FpLq6mnsDg6Mt
YiXcl8nlVj682vHY9AubX8L9HeSYDmX7fe9SdrN1W16AUAi0Ecq4OtlVIlbMk1EC
DQKWib31LOgN8kI1MdTsYhukEK2rJcZeHHaSOWeZgD8XdTNyxesDFt0DmQZ/T2HO
3gMo+xPVUczESTC1ltFo80MmJterPNWYJRZL6JN22tH+WNbMq8P6A0X4ZyP7DU9A
Ntj4bNj18z7ykedp6H0lmzLoM1F+3PRMaq8LCBFwB7rwQmUu2OC9cfC9+bHY5wad
AUwFx4HHmD/UCcxaIjMYKXJ6di8WP1LqHlPxyeCGKUCT8MH1Xqv/1NcO2/KqFedn
MZKS9beJyw6wVQMFBMiR3+08IEdoxLAuoyC3tTKw7zQ8NxQdi5bS7N+aAlfXcsLF
cl4Zhp/e88IUDAC57MLqXlv1WFf/BR4dA0FPHG20ZMyDrqzb4swhJX1K9IIGO+3L
1j3jx38A5r/2kZlJs+g/CFk0+RhpcXQbQVXXl+v+VOCLMhtb4gtqTsBhIJLZ+sWw
```

Dans un second temps je vais créer un certificat auto-signé (durée de vie de 10 ans)

openssl req -x509 -new -nodes -key ca.key.pem -sha256 -days 3650 -out ca.cert.pem -subj "/C=FR/ST=France/L=Paris/O=esgialexis/OU=esgialexis/CN=Root CA"
chmod 444 ca.cert.pem

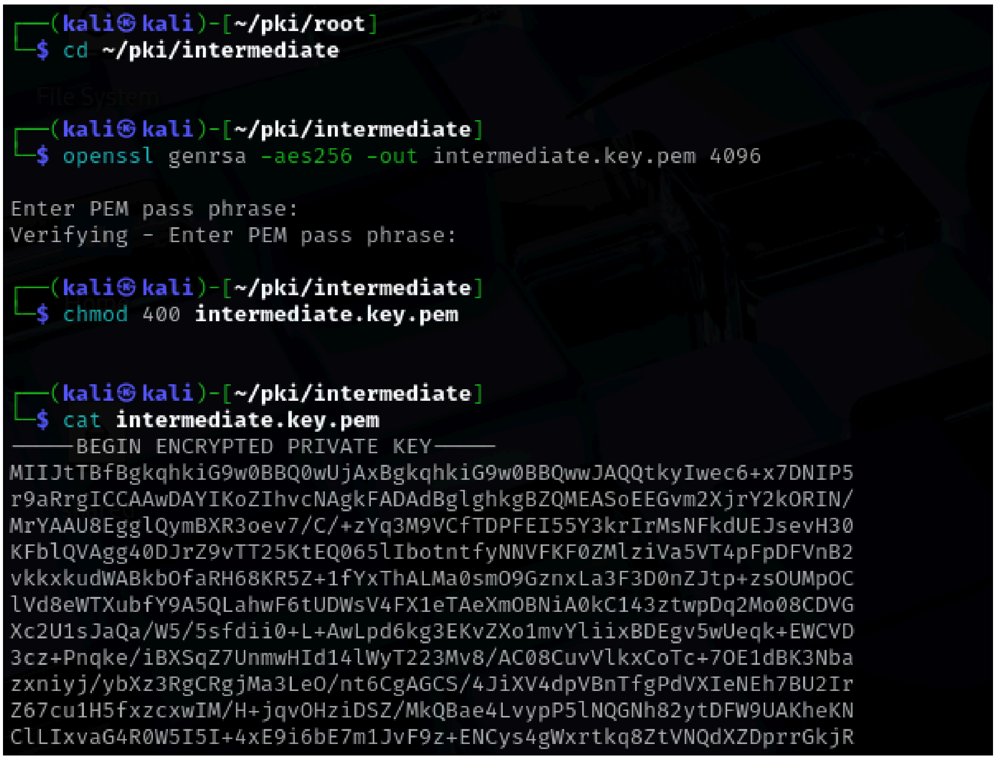
```
┌──(kali㉿kali)-[~/pki/root]
└─$ openssl req -x509 -new -nodes -key ca.key.pem -sha256 -days 3650 -out ca.cert.pem -subj "/C=FR/ST=France/L=Paris/O=esgialexis/OU=esgialexis/CN=Root CA"


Enter pass phrase for ca.key.pem:

┌──(kali㉿kali)-[~/pki/root]
└─$ ls
ca.cert.pem  ca.key.pem

┌──(kali㉿kali)-[~/pki/root]
└─$ cat ca.cert.pem
-----BEGIN CERTIFICATE-----
MIIFtTCCA52gAwIBAgIUN5GC5nOs1jHv9V10bPJ3xOKSMXswDQYJKoZIhvcNAQEL
BQAwajELMAkGA1UEBhMCRlIxDzANBgNVBAgMBkZyYW5jZTEOMAwGA1UEBwwFUGFy
aXMxEzARBgNVBAoMCmVzZ2lhbGV4aXMxEzARBgNVBAsMCmVzZ2lhbGV4aXMxEDAO
BgNVBAMMB1Jvb3QgQ0EwHhcNMjUwNTIwMDk1MDI5WhcNMzUwNTE4MDk1MDI5WjBq
MQswCQYDVQQGEwJGUjEPMA0GA1UECAwGRnJhbmNlMQ4wDAYDVQQHDAVQYXJpczET
MBEGA1UECgwKZXNnaWFsZXhpczETMBEGA1UECwwKZXNnaWFsZXhpczEQMA4GA1UE
AwwHUm9vdCBDQTCCAiIwDQYJKoZIhvcNAQEBBQADggIPADCCAgoCggIBALNX7G2+
zuOousNS2/tofNZXbbyVZZxNs6Qj52z26b5YWirAw/DAVxo37zdsxzxYTvZYA0Rh
LtYro0VmlaxBxVRkRfCrpU/RqshTJ8334aeIaghsvoAmCljdcJfr2K3OC2Kj8sOm
dayJ463TWaIvGXNydS/NVHOJWq5WMet8LtHoRwMeXIpJc9Ye1kg5nYoV7xXPldGU
+7GNaz/k5AaMS8Y1hjBvp7EEhrYP8ByUch7kj+eP873pMbqwW+Rf8Txy9oOkJttu
```

Alexis Brunel 21/05/2025

Création de CA intermédiaire :

la clé privé pour la CA intermédiaire :

openssl genrsa -aes256 -out intermediate.key.pem 4096



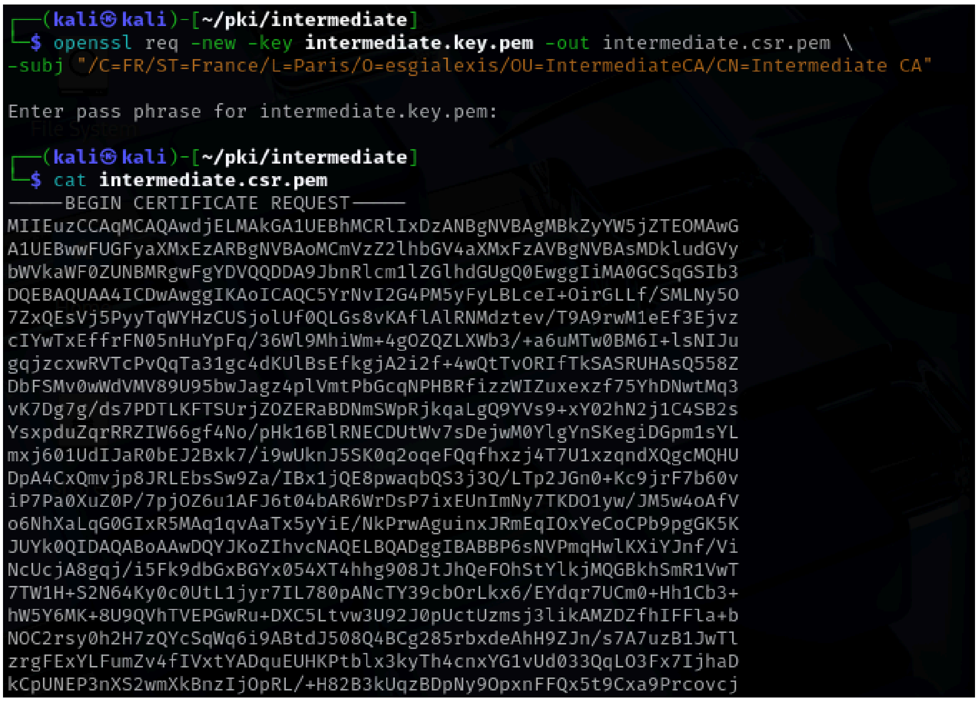
```
┌──(kali㉿kali)-[~/pki/root]
└─$ cd ~/pki/intermediate

┌──(kali㉿kali)-[~/pki/intermediate]
└─$ openssl genrsa -aes256 -out intermediate.key.pem 4096

Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:

┌──(kali㉿kali)-[~/pki/intermediate]
└─$ chmod 400 intermediate.key.pem

┌──(kali㉿kali)-[~/pki/intermediate]
└─$ cat intermediate.key.pem
-----BEGIN ENCRYPTED PRIVATE KEY-----
```


```
MIIJtTBfBgkqhkiG9w0BBQ0wUjAxBgkqhkiG9w0BBQwwJAQQtkyIwec6+x7DNIP5
r9aRrgICCAAwDAYIKoZIhvcNAgkFADAdBglghkgBZQMEASoEEGvm2XjrY2kORIN/
MrYAAU8EgglQymBXR3oev7/C/+zYq3M9VCfTDPFEI55Y3krIrMsNFkdUEJsevH30
KFblQVAgg40DJrZ9vTT25KtEQ065lIbotntfyNNVFKF0ZMlziVa5VT4pFpDFVnB2
vkkxkudWABkbOfaRH68KR5Z+1fYxThALMa0smO9GznxLa3F3D0nZJtp+zsOUMpOC
lVd8eWTXubfY9A5QLahwF6tUDWsV4FX1eTAeXmOBNiA0kC143ztwpDq2Mo08CDVG
Xc2U1sJaQa/W5/5sfdii0+L+AwLpd6kg3EKvZXo1mvYliixBDEgv5wUeqk+EWCVD
3cz+Pnqke/iBXSqZ7UnmwHId14lWyT223Mv8/AC08CuvVlkxCoTc+7OE1dBK3Nba
zxniyj/ybXz3RgCRgjMa3LeO/nt6CgAGCS/4JiXV4dpVBnTfgPdVXIeNEh7BU2Ir
Z67cu1H5fxzcxwIM/H+jqvOHziDSZ/MkQBae4LvypP5lNQGNh82ytDFW9UAKheKN
ClLIxvaG4R0W5I5I+4xE9i6bE7m1JvF9z+ENCys4gWxrtkq8ZtVNQdXZDprrGkjR
```


J'utilise la passphrase **alexisint**

Création d'un certificat

openssl req -new -key intermediate.key.pem -out intermediate.csr.pem \ -subj "/C=FR/ST=France/L=Paris/O=esgialexis/OU=IntermediateCA/CN=Intermediate CA"



```
┌──(kali㉿kali)-[~/pki/intermediate]
└─$ openssl req -new -key intermediate.key.pem -out intermediate.csr.pem \
-subj "/C=FR/ST=France/L=Paris/O=esgialexis/OU=IntermediateCA/CN=Intermediate CA"

Enter pass phrase for intermediate.key.pem:

┌──(kali㉿kali)-[~/pki/intermediate]
└─$ cat intermediate.csr.pem
-----BEGIN CERTIFICATE REQUEST-----
```


```
MIIEuzCCAqMCAQAwdjELMAkGA1UEBhMCRlIxDzANBgNVBAgMBkZyYW5jZTEOMAwG
A1UEBwwFUGFyaXMxEzARBgNVBAoMCmVzZ2lhbGV4aXMxFzAVBgNVBAsMDkludGVy
bWVkaWF0ZUNBMRgwFgYDVQQDDA9JbnRlcm1lZGlhdGUgQ0EwggIiMA0GCSqGSIb3
DQEBAQUAA4ICDwAwggIKAoICAQC5YrNvI2G4PM5yFyLBLceI+0irGLLf/SMLNy5O
7ZxQEsVj5PyyTqWYHzCUSjolUf0QLGs8vKAflAlRNMdztev/T9A9rwM1eEf3Ejvz
cIYwTxEffrFN05nHuYpFq/36Wl9MhiWm+4gOZQZLXWb3/+a6uMTw0BM6I+lsNIJu
gqjzcxwRVTcPvQqTa31gc4dKUlBsEfkgjA2i2f+4wQtTvORIfTkSASRUHAsQ558Z
DbFSMv0wWdVMV89U95bwJagz4plVmtPbGcqNPHBRfizzWIZuxexzf75YhDNwtMq3
vK7Dg7g/ds7PDTLKFTSUrjZOZERaBDNmSWpRjkqaLgQ9YVs9+xY02hN2j1C4SB2s
YsxpduZqrRRZIW66gf4No/pHk16BlRNECDUtWv7sDejwM0YlgYnSKegiDGpm1sYL
mxj601UdIJaR0bEJ2Bxk7/i9wUknJ5SK0q2oqeFQqfhxzj4T7U1xzqndXQgcMQHU
DpA4CxQmvjp8JRLEbsSw9Za/IBx1jQE8pwaqbQS3j3Q/LTp2JGn0+Kc9jrF7b60v
iP7Pa0XuZ0P/7pjOZ6u1AFJ6t04bAR6WrDsP7ixEUnImNy7TKDO1yw/JM5w4oAfV
o6NhXaLqG0GIxR5MAq1qvAaTx5yYiE/NkPrwAguinxJRmEqIOxYeCoCPb9pgGK5K
JUYk0QIDAQABoAAwDQYJKoZIhvcNAQELBQADggIBABBP6sNVPmqHwlKXiYJnf/Vi
NcUcjA8gqj/i5Fk9dbGxBGYx054XT4hhg908JtJhQeFOhStYlkjMQGBkhSmR1VwT
7TW1H+S2N64Ky0c0UtL1jyr7IL780pANcTY39cbOrLkx6/EYdqr7UCm0+Hh1Cb3+
hW5Y6MK+8U9QVhTVEPGwRu+DXC5Ltvw3U92J0pUctUzmsj3likAMZDZfhIFFla+b
NOC2rsy0h2H7zQYcSqWq6i9ABtdJ508Q4BCg285rbxdeAhH9ZJn/s7A7uzB1JwTl
zrgFExYLFumZv4fIVxtYADquEUHKPtblx3kyTh4cnxYG1vUd033QqLO3Fx7IjhaD
kCpUNEP3nXS2wmXkBnzIjOpRL/+H82B3kUqzBDpNy9OpxnFFQx5t9Cxa9Prcovcj
```

Signature du certificat de l'intermédiaire par la racine :

Création d'un fichier
nano intext.cnf
→ basicConstraints = critical, CA:TRUE, pathlen:0
keyUsage = critical, cRLSign, keyCertSign

**openssl x509 -req **
**-in ../intermediate/intermediate.csr.pem **
**-CA ca.cert.pem -CAkey ca.key.pem -CAcreateserial **
**-out ../intermediate/intermediate.cert.pem -days 1825 -sha256 **
-extfile intext.cnf

```
┌──(kali㉿kali)-[~/pki/root]
└─$ nano intext.cnf

┌──(kali㉿kali)-[~/pki/root]
└─$ openssl x509 -req \
-in ../intermediate/intermediate.csr.pem \
-CA ca.cert.pem -CAkey ca.key.pem -CAcreateserial \
-out ../intermediate/intermediate.cert.pem -days 1825 -sha256 \
-extfile intext.cnf

Certificate request self-signature ok
subject=C=FR, ST=France, L=Paris, O=esgialexis, OU=IntermediateCA, CN=Intermediate CA
Enter pass phrase for ca.key.pem:

┌──(kali㉿kali)-[~/pki/root]
└─$ cat intext.cnf
basicConstraints = critical, CA:TRUE, pathlen:0
keyUsage = critical, cRLSign, keyCertSign
```

Petite vérification

```
┌──(kali㉿kali)-[~/pki/root]
└─$ openssl x509 -in ../intermediate/intermediate.cert.pem -noout -text | grep -A10 "X509v3 Basic Constraints"
            X509v3 Basic Constraints: critical
                CA:TRUE, pathlen:0
            X509v3 Key Usage: critical
                Certificate Sign, CRL Sign
            X509v3 Subject Key Identifier:
                92:17:FA:24:47:58:D5:80:4B:73:DA:E9:51:3F:42:01:70:1A:0B:50
            X509v3 Authority Key Identifier:
                AE:57:03:D9:EE:BA:79:B7:79:48:DE:11:A5:DD:9C:33:B1:12:63:E9
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        1c:16:ad:37:dc:f4:1c:71:65:bf:37:08:aa:38:71:44:8c:87:
```

Vérification rapide de ce qui est bon :

- **✅ CA:TRUE, pathlen:0 dans X509v3 Basic Constraints → autorisée à signer, mais pas de CA en dessous (c'est bien ce qu'on veut)**

- **✅ Key Usage: Certificate Sign, CRL Sign → autorisée à signer des certificats et publier des CRL**

- **✅ Signée avec sha256WithRSAEncryption**

- **✅ Subject et Authority Key Identifier sont bien définis**

Configuration coté serveur

Génération d'une clé et d'un CSR avec comme serveur le localhost.

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ openssl genrsa -out server.key.pem 2048

┌──(kali㉿kali)-[~/pki/certs]
└─$ openssl req -new -key server.key.pem -out server.csr.pem \
-subj "/C=FR/ST=France/L=Paris/O=esgialexis/OU=WebServer/CN=localhost"
```

Demande de signature par la CA intermédiaire :

nano servtxt.cnf

→ **basicConstraints = CA:FALSE**
keyUsage = critical, Digital Signature, Key Encipherment
extendedKeyUsage = serverAuth
subjectAltName = @alt_names

[alt_names]
DNS.1 = localhost

**openssl x509 -req -in server.csr.pem -CA ../intermediate/intermediate.cert.pem **
**-CAkey ../intermediate/intermediate.key.pem -CAcreateserial **
-out server.cert.pem -days 365 -sha256 -extfile servtxt.cnf

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ openssl x509 -req -in server.csr.pem -CA ../intermediate/intermediate.cert.pem \
-CAkey ../intermediate/intermediate.key.pem -CAcreateserial \
-out server.cert.pem -days 365 -sha256 -extfile servtxt.cnf

Certificate request self-signature ok
subject=C=FR, ST=France, L=Paris, O=esgialexis, OU=WebServer, CN=localhost
Enter pass phrase for ../intermediate/intermediate.key.pem:
```

Puis je vais créer la full chain :

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ cat server.cert.pem ../intermediate/intermediate.cert.pem ../root/ca.cert.pem > fullchain.pem
```

Donc maintenant j'ai précisément :

server.key.pem → clé privée du serveur

server.cert.pem → certificat signé par la CA intermédiaire

fullchain.pem → certificat + intermédiaire + racine pour Apache

Je vais préparer l'environnement pour apache2

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo mkdir -p /etc/ssl/esgialexis
sudo cp ~/pki/certs/server.key.pem /etc/ssl/esgialexis/
sudo cp ~/pki/certs/server.cert.pem /etc/ssl/esgialexis/
sudo cp ~/pki/certs/fullchain.pem /etc/ssl/esgialexis/
```

Puis j'installe apache et j'active ssl :

sudo apt install apache2
sudo a2enmod ssl

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo apt install apache2

[sudo] password for kali:
apache2 is already the newest version (2.4.63-1).
apache2 set to manually installed.
The following packages were automatically installed and are no longer required:
  crackmapexec            libegl-dev          libgtksourceview-3.0-1          libnetcdf19t64          libtag1v5-vanilla
  firebird3.0-common      libflac12t64        libgtksourceview-3.0-common     libpaper1               libtagc0
  firebird3.0-common-doc  libfuse3-3          libgtksourceviewmm-3.0-0v5      libpoppler140           libunwind-19
  icu-devtools            libgdal35           libgumbo2                       libpoppler145           libwebrtc-audio-proces
  libapt-pkg6.0t64        libgeos3.13.0       libhdf5-103-1t64                libpython3.12-dev       libx265-209
  libc++1-19              libgl1-mesa-dev     libhdf5-hl-100t64               libpython3.12-minimal   linux-image-6.6.9-amd6
  libc++abi1-19           libglapi-mesa       libicu-dev                      libpython3.12-stdlib    openjdk-23-jre
  libcapstone4            libgles-dev         libjxl0.9                       libpython3.12t64        openjdk-23-jre-headles
  libconfig++9v5          libgles1            liblbfgsb0                      libqt5sensors5          python3-appdirs
  libconfig9              libglvnd-core-dev   libldap-2.5-0                   libqt5webkit5           python3-ntlm-auth
  libdirectfb-1.7-7t64    libglvnd-dev        libmsgraph-0-1                  libtag1v5               python3-setproctitle
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 13

┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo a2enmod ssl

Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
```

Je met par défaut le ssl :

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo a2ensite default-ssl.conf

Enabling site default-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
```

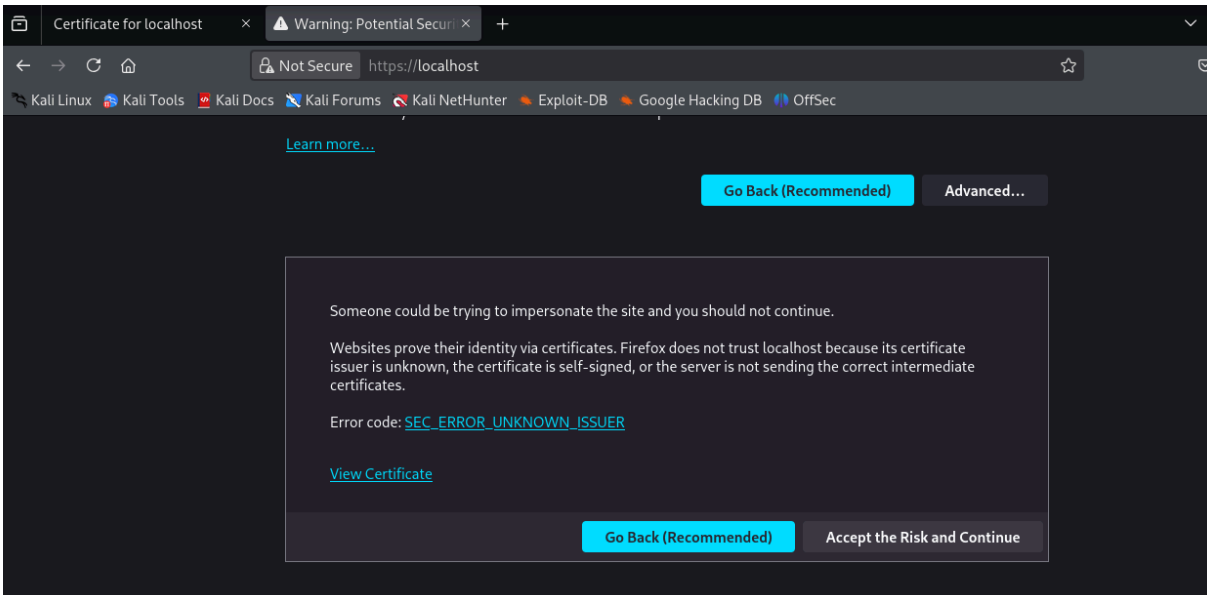
Je vais modifier la configuration apache2 pour pointer vers mes certificats :

```
kali@kali: ~/pki/certs
File  Actions  Edit  View  Help
 GNU nano 8.4                    /etc/apache2/sites-available/default-ssl.conf
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf

        #   SSL Engine Switch:
        #   Enable/Disable SSL for this virtual host.
        SSLEngine on

        #   A self-signed (snakeoil) certificate can be created by installing
        #   the ssl-cert package. See
        #   /usr/share/doc/apache2/README.Debian.gz for more info.
        #   If both key and certificate are stored in the same file, only the
        #   SSLCertificateFile directive is needed.
        SSLCertificateFile      /etc/ssl/esgialexis/server.cert.pem
        SSLCertificateKeyFile   /etc/ssl/esgialexis/server.key.pem
        SSLCertificateChainFile /etc/ssl/esgialexis/fullchain.pem
        #   Server Certificate Chain:
        #   Point SSLCertificateChainFile at a file containing the
        #   concatenation of PEM encoded CA certificates which form the
        #   certificate chain for the server certificate. Alternatively
        #   the referenced file can be the same as SSLCertificateFile
        #   when the CA certificates are directly appended to the server

^G Help      ^O Write Out  ^F Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark  M-] To Bracket  M-B Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy      ^B Where Was    M-F Next
```

Je redémarre apache2

```
┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo nano /etc/apache2/sites-available/default-ssl.conf

┌──(kali㉿kali)-[~/pki/certs]
└─$ sudo systemctl restart apache2
```

Je vais vérifier dans le navigateur : https://localhost

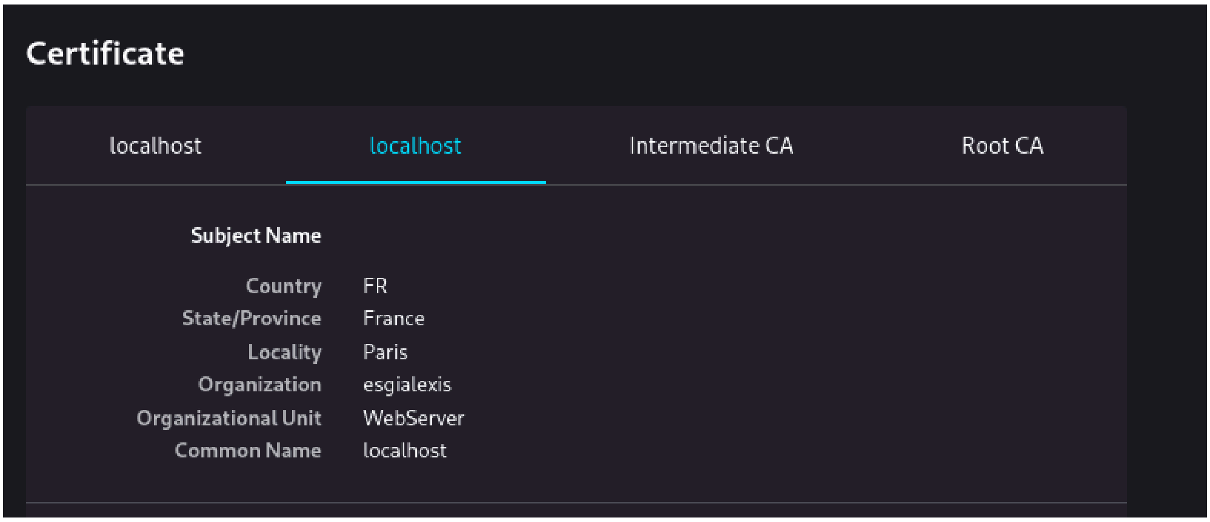


On retrouve ce message car :

- Mon navigateur ne connaît pas ta Root CA → il ne peut donc pas faire confiance à ma chaîne de certification.
- Mon certificat n'est pas signé par une autorité publique reconnue (genre Let's Encrypt, DigiCert, etc.).

Je clique sur view certificat :

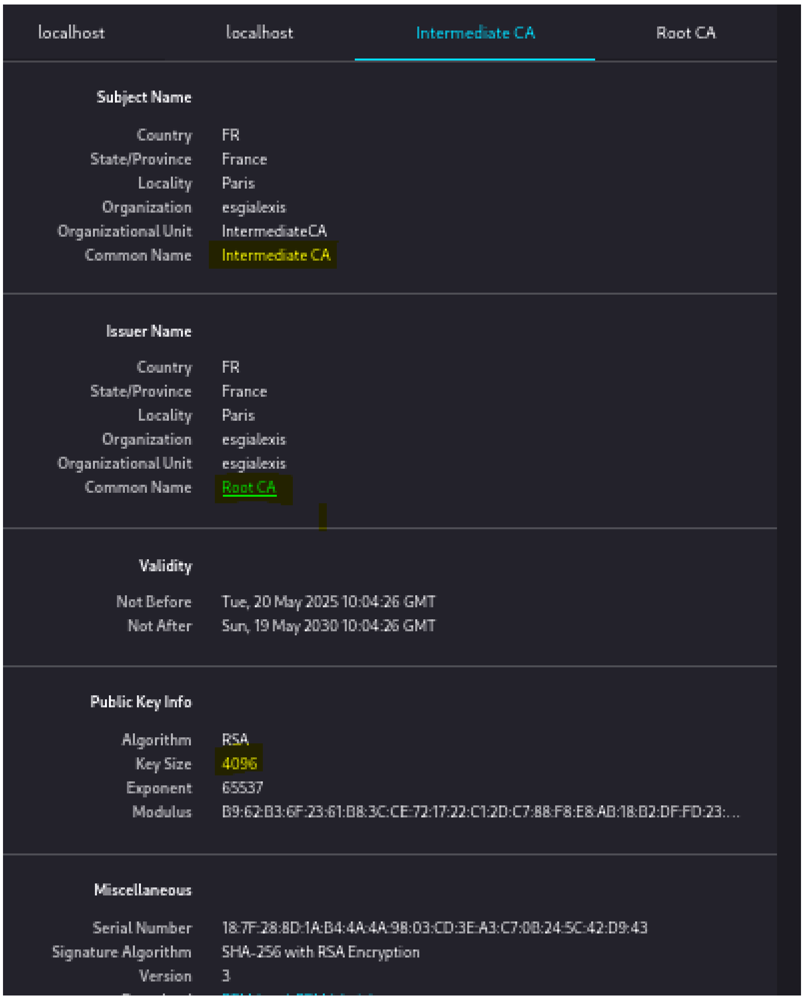
Je retrouve différents onglet :



Dans localhost je suis censé trouver le certificat de mon serveur apache.

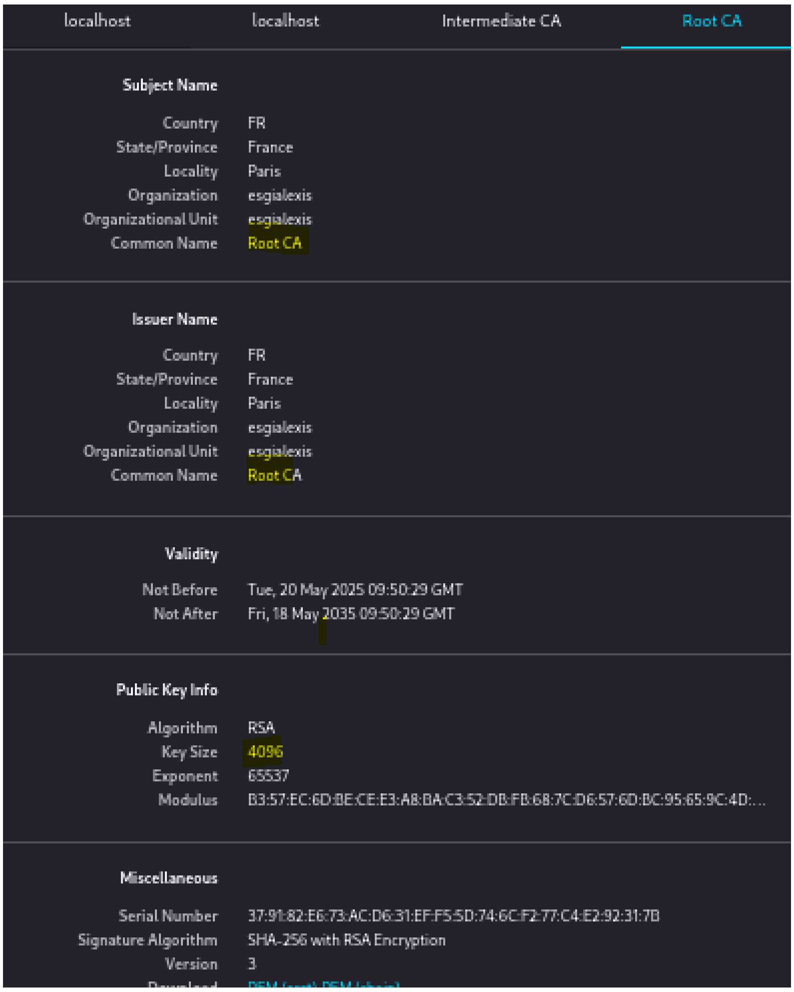
Il est bien de 2048 et pas 4096 comme les 2 autres, il est donné par le “intermediateCA”

Dans l’onglet “intermediateCA” on retrouve



On remarque qu’il est intermédiaire étant donné que c’est le RootCA qui délivre le certificat et que le serveur à demandé au “intermediateCA” pour son certificat.

Puis au niveau de du RootCA il s’agit d’un certificat classique auto-signé :



Voici à quoi ressemble le certificat serveur :